SIMATS ENGINEERING
ASSESSMENT TOOL 3

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks.* *(BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

TECHNICAL PRESENTATION

Code of Conduct:

I D. Sheshuraaj | 196526002 *(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.*
I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student:

Criteria	Weightage	Q1 30marks
Understanding of Concepts	25	
Experimental Design	30	
Use of Tools	20	
Analysis of Results	15	
Documentation	10	

Total Marks 25 **/30**

Signature of the Faculty

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SIMATS

Saveetha Institute of Medical And Technical Sciences

(Declared as Deemed to be University under Section 3 of UGC Act 1956)

Machine Learning – Based Error Detection in Networks

CO1-AT3 TECHNICAL PRESENTATION

CSA0709 COMPUTER NETWORKS

GUIDED BY

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ABSTRACT

- Network security is important for protecting systems and data from cyber attacks.
- Network traffic contains valuable information about communication between devices.
- Monitoring traffic helps identify abnormal and suspicious activities.
- Traditional security methods may not detect new threats quickly.
- This project provides real-time monitoring, threat identification, and alerts to improve network security.

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INTRODUCTION

- Real-time monitoring of network traffic and data packets.
- Collects network information such as IP addresses, ports, protocols, and packet size.
- Detects suspicious activities and abnormal network behavior.
- Generates real-time alerts when cyber threats are identified.
- Provides a dashboard to visualize traffic, threats, severity, and security status.

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PROBLEM STATEMENT

- Increasing cyber attacks make network security difficult to maintain.
- Suspicious network activities may go unnoticed without continuous monitoring.
- Large volumes of network traffic are difficult to analyze manually.
- Delayed threat detection can cause data loss and system damage.
- A real-time automated system is needed to detect threats and provide immediate alerts.

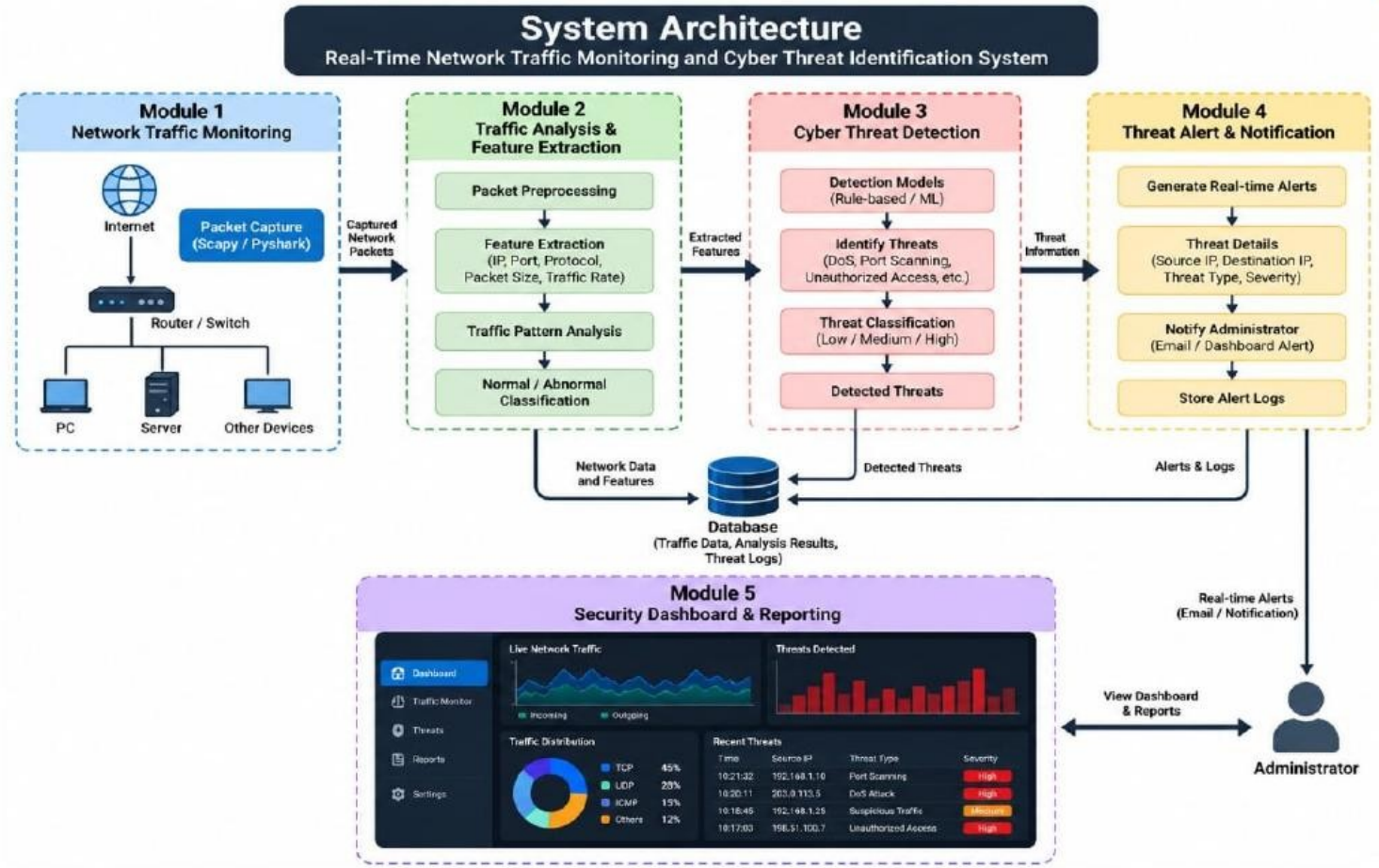
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OBJECTIVES

- Monitor network traffic continuously in real time.
- Capture and analyze network packets and traffic information.
- Identify suspicious and abnormal network activities.
- Detect cyber threats quickly and generate real-time alerts.
- Provide a clear dashboard for monitoring traffic and detected threats.

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SYSTEM ARCHITECTURE



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MODULE 1 – Network Traffic Monitoring

- Capture network traffic from connected devices in real time.
- Collect details such as IP address, port, protocol, and packet size.
- Monitor incoming and outgoing network packets continuously.
- Analyze traffic patterns to identify normal and unusual behavior.
- Display live traffic information on the monitoring dashboard.

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MODULE 2 –TRAFFIC ANALYSIS & FEATURE EXTRACTION

- Analyze the captured network traffic.
- Extract important features such as IP, port, protocol, packet size, and traffic rate.
- Identify normal and abnormal traffic patterns.
- Prepare the extracted data for threat detection.
- Send analyzed data to the cyber threat detection module.

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MODULE 3 –CYBER THREAT DETECTION

- Detect suspicious and abnormal network activities.
- Identify threats such as DoS attacks, port scanning, and unauthorized access.
- Compare traffic patterns with security rules and detection models.
- Classify the detected activity based on threat type and severity.
- Send detected threats to the alert and response module.

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MODULE 4–THREAT ALERT & NOTIFICATION

- Generate real-time alerts when a cyber threat is detected.
- Display the threat type, source IP, destination IP, and severity.
- Assign threat levels such as Low, Medium, and High.
- Notify the administrator about critical network activities.
- Store alert details for future analysis and reporting.

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MODULE 5 - SECURITY DASHBOARD & REPORTING

- Display real-time network traffic through a user-friendly dashboard.
- Show detected threats and their severity levels.
- Provide charts for traffic patterns and security events.
- Store and display historical threat and network data.
- Generate security reports for analysis and future monitoring

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ADVANTAGES & DISADVANTAGES

- Real-time monitoring of network traffic.
- Early threat detection and immediate alerts.
- Reduces manual efforts in monitoring large networks.
- Provides a dashboard for security analysis.
- Helps improve overall network security.
- Requires continuous system updates.
- Can generate false alerts for unusual but safe activities.
- High resource consumption.
- Initial setup and configuration can be complex.
- Helps in recovery and learning from past incidents.
- Enhances the accuracy of detection models used

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CONCLUSION

The system provides real-time monitoring of network traffic.It helps identify suspicious activities and cyber threats quickly.Real-time alerts and notifications improve security response.The dashboard provides clear visualization and analysis of network activities.Overall, the system helps achieve faster and more effective network security.

REFERENCES

- William Stallings, Network Security Essentials: Applications and Standards, 7th Edition, 2017.
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- NIST, Cybersecurity Framework (CSF) 2.0, 2024.
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- Scapy Documentation, Scapy: Packet Manipulation Program, 2025.


Thank
you